GHAYDA NA'IM JA'AFREH

Data Scientist | Python | AI Model Evaluation | Machine Learning | Technical Reasoning

Karak, Jordan | ghayda.jaafreh@gmail.com

linkedin.com/in/ghayda-njaafreh | orcid.org/0009-0009-4853-8874 | github.com/ghayda-njaafreh | kaggle.com/jaafreh

PROFESSIONAL SUMMARY

Data Science and Artificial Intelligence graduate with an Excellent GPA of 93.04%, ranked second in the graduating cohort. Hands-on experience developing, debugging, evaluating, and documenting Python-based AI and data-science workflows. Strong in model benchmarking, comparative experimentation, error analysis, data preparation, threshold evaluation, and clear technical reasoning. Built end-to-end projects spanning healthcare AI, computer vision, autonomous gameplay, multimodal systems, and reproducible experiments.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, data cleaning, preprocessing, exploratory analysis, feature engineering, structured datasets
Machine Learning & AI: Classification, regression, predictive modeling, deep learning, CNNs, transfer learning, segmentation, explainable AI, model inference
Evaluation & Benchmarking: Accuracy, precision, recall, F1, ROC-AUC, PR-AUC, confusion matrices, Dice, IoU, mAP, learning curves, threshold evaluation, calibration, error analysis
Frameworks & Libraries: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Matplotlib, Gradio, FastAPI
Engineering & Reasoning: Debugging, code analysis, input validation, logging, Git/GitHub, reproducible experimentation, technical documentation, rationale writing
Tools & Platforms: Jupyter Notebook, Azure-hosted GPU environment, SQLite, MySQL, Excel, Tableau, Kaggle

EDUCATION

Bachelor's Degree in Data Science and Artificial Intelligence | Mutah University | 2022-2026
GPA: 93.04% | Grade: Excellent | Ranked 2nd in the Data Science and Artificial Intelligence graduating cohort

SELECTED PROJECTS

StrokeLens - Secure & Explainable Healthcare AI Graduation Project

Graduation Project | Sep 2025-Jun 2026 | Core Project Owner and Lead Technical Contributor

- Led an end-to-end healthcare AI project using structured clinical data, multi-stage stroke-risk modeling, explainability, secure reporting, and integrated application workflows.
- Contributed across data preparation, model experimentation and evaluation, threshold analysis, explainability, system integration, testing, technical documentation, and Azure-hosted GPU experimentation.

Technologies: Python, machine learning, explainable AI, model evaluation, secure systems, Azure GPU

Tetris AI Lab - End-to-End AI Game Application

Individual Project | Public GitHub Release

- Developed and published a Python/Pygame application integrating a CNN policy with inference, top-5 one-piece lookahead, safety controls, replay functionality, and human/AI gameplay modes.
- Built a reproducible training-data pipeline with automated cleaning, video-level train/validation/test splitting, simulator-generated recovery states, simulator-verified mirror augmentation, experiment tracking, testing, and release documentation.

Technologies: Python, PyTorch, CNN, Pygame, data pipelines, testing, reproducible experimentation

MNIST CNN Batch Normalization Comparison

Kaggle Deep Learning Project | Comparative Model Evaluation

- Implemented two PyTorch CNN variants under controlled conditions and compared accuracy, loss, runtime, classification reports, and confusion matrices; achieved 99.37% test accuracy with Batch Normalization.

Technologies: Python, PyTorch, CNN, Batch Normalization, model evaluation

Medical Image Segmentation & Computer Vision Portfolio

Kaggle and Academic Projects

- Built teeth instance segmentation for dental X-rays, achieving Mask mAP50 of 0.9947 and Mask mAP50-95 of 0.6301.
- Implemented a custom PyTorch U-Net workflow for dental-caries segmentation, achieving 0.9010 test Dice; additional work includes Intel transfer learning, MNIST CNN experiments, OCR with HOG/KNN, and image classification.

Technologies: Python, PyTorch, OpenCV, U-Net, transfer learning, OCR, segmentation

ML Decision Surfaces Lab - Interactive Machine Learning Playground

Team Project | Main Algorithm Developer and Interface Designer

- Built an interactive Gradio application for comparing classification and regression models using decision boundaries, ROC-AUC, confusion matrices, learning curves, and controlled noise/outlier experiments.

Technologies: Python, scikit-learn, Gradio, Matplotlib, model evaluation

Privacy-Safe Multimodal Authentication System

Team Project | Main Voice-Model and Integration Contributor

- Led work on the voice-verification component and contributed to integration, verification logic, API-related workflow, testing, logging, and privacy-aware data handling.
- Integrated voice verification with a Python/FastAPI multimodal workflow that also incorporated face verification and optional offline speech recognition; resolved runtime, dependency, and input-format issues through systematic testing.

Technologies: Python, FastAPI, PyTorch, SpeechBrain, OpenCV, InsightFace, Vosk, REST APIs, authentication, logging

EXPERIENCE

Data Science & Programming Trainee | The Hope International Company | Online Training | Jun 20, 2025-Sep 10, 2025

- Completed structured practical training in Python programming, data analysis, preprocessing, visualization, model implementation, debugging, teamwork, and technical documentation.
- Applied analytical and programming skills to structured datasets and small end-to-end AI workflows.

AI & Machine Learning Trainee | The Hope International Company | Online Training | Nov 15, 2023-Mar 15, 2024

- Built and evaluated Python-based machine-learning exercises covering data preparation, feature selection, classification, regression, decision trees, KNN, Naive Bayes, and model evaluation.

- Developed end-to-end understanding of the machine-learning lifecycle from preparation through evaluation and reporting.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft | Earned May 27, 2026 | Credential ID: C5834AFC20D9A41C | Certification No.: B010AT-7276E5
- Data Science Essentials with Python | Cisco Networking Academy | May 29, 2026 | Certificate ID: 960f2b70-a252-42a1-a73d-13e7ca5c7c51
- Python Essentials 1 | Cisco Networking Academy | May 30, 2026 | Certificate ID: 01720f08-f6be-48a8-b730-d4da54339d1f
- Deep Learning - 60 Hours | The Hope International Company | Sep 15, 2025 | Certificate ID: 1758145099980
- Python AI (Machine Learning) - 60 Hours | The Hope International Company | Nov 2023-Mar 2024
- Ethical Hacker | Cisco Networking Academy | May 29, 2026 | Certificate ID: 34898e2b-05cf-41a1-bcc0-c8d3f49090d1
- Cyber Security Associate - Certificate of Completion | Green Circle for Software Solutions | 2026
- CCNA Course Certificate | Pioneers Academy | Dec 28, 2023 | Certificate No.: 18437657-24
- Security+ Course Certificate - 30 Hours | Pioneers Academy | Oct 5, 2023 | Certificate No.: 18424325-2560
- Human Research: Data or Specimens Only Research - Basic Course | CITI Program, MIT Affiliates | Jul 13, 2025-Jul 13, 2028 | Record ID: 70859362
- HTML Essentials | Cisco Networking Academy | May 30, 2026 | Certificate ID: 1ca22c8d-bbae-4328-8cf8-07eacac17e5e
- International Computer Driving License (ICDL) - 36 Hours | Arab International Academy | Feb 5, 2024 | CRT.No: 201548
- English Communication Skills Training - 20 Hours | Mutah University / King Abdullah II Fund for Development | Apr 6-May 13, 2024
- English Conversation - 12 Levels, 90 Hours | Arab International Academy | Feb 1-May 15, 2023 | ID: 200166016 | RN: 117840

LANGUAGES & CAREER FOCUS

Languages: Arabic (Native), English (Intermediate)
Career Focus: Data Science, AI Model Evaluation, Machine Learning, Technical Reasoning, Data Quality, Experimentation, and Applied AI Research.